Project Overview

Project Context

Role:

- You should act as a consultant for eSC, an energy company

Goal:

- How might your client predict future energy usage?
  - Predict energy usage if the summer was 5 degrees warmer
  - Provide actionable insight into how to reduce energy costs

What data to analyze – should be:

- A function of what the team determines might be useful
- Determined by each project team
- There is *A LOT* of data

Remember this needs to be data driven

Project Data

Static House Data

A file with basic house information for a random sample of single-family houses that eSC serves.

- The file contains the list of all houses in the dataset.
- For each house, there is information describing the house.
  - The information ranges from the building id (used to access the energy data) to other house attributes that do not change (such as the size of the house).
- There are >5,000 houses in the dataset (rows in the file)
- The file can be found at: https://intro-datascience.s3.us-east-2.amazonaws.com/SC-data/static_house_info.parquet

Note that this file is in 'parquet' format an optimized binary format

Energy Usage Data

- For each house, there is a file that contains energy usage data, which was collected hour-by-hour.

- There is one data file per house. Energy usage is:
  - Collected every hour
  - Collected across many sources (ex. air conditioning system, dryer)
  - the 'building ID' is file name which identifies the house.

- Note that each file is in 'parquet' format.

- All the data is in one folder on Amazon AWS.

- For example, the following URL is for 'building_id' 102063.

  https://intro-datascience.s3.us-east-2.amazonaws.com/SC-data/2023-houseData/102063.parquet

There are more than 5,000 files in the directory (one for each building id).

Weather Data

- Hour-by-hour weather information
(one file for each geographic area)

- The weather data was collected for each county and stored based on a county code:
  - The county code for each house can be found at 'in.county' column of the house static dataset. This file is in a simple CSV format.
  - For example, the following URL provides the weather for county ‘G4500010’.

    https://intro-datascience.s3.us-east-2.amazonaws.com/SC-data/weather/2023-weather-data/G4500010.csv

There are approximately 50 files in the directory (one for each county).

Meta Data

- A data description file, explaining the fields used across the different housing data files.

  ~270 attributes

  https://intro-datascience.s3.us-east-2.amazonaws.com/SC-data/data_dictionary.csv

Project Approach

Suggested Project Steps

a) Determine the best approach to read and merge the data.

b) Determine what should be the output during this 'data preparation' phase.

b) Do exploratory analysis of the data.

c) Build a model that predicts the energy usage, for any given hour, for the month of July.
→ July was selected, as eSC thought July is typically the highest energy usage month.
Hint: you will need to try several models and pick the best model.

d) Understand and be able to explain your model's accuracy.

Suggested Project Steps

e) Create a new weather dataset → all July temps 5 degrees warmer.

f) Use your best model to evaluate peak future energy demand.
→ Assume no new customers
Note: this must be model driven, not just increasing energy usage by a percentage

g) Show future peak energy demand in total (for an hour):
- For different geographic regions
- For other dimensions /attributes you think important

h) Create a shiny application to interact with the data.

i) Identify one approach to reduce peak energy demand.

j) What would you suggest, how would you model the impact?

k) How would you explain the impact? BE DATA DRIVEN!

Project Deliverables

Project Deliverables

Word Document:

- Target audience is your manager / instructor
  (hint: your manager/instructor is a data science expert)
- Focus on what was accomplished
- Should describe all analysis done, even if an analysis did not generate any interesting results, it should still be included
- This should be *expository* in nature

Presentation:

- This should be high-level
- Target audience is your client *(hint: the client is not a data science expert)*
- Presentation length is 8-10 minutes *(lab instructor will explain specifics)*
- Be sure to include the key drivers of every usage, model results and accuracy, and recommendations

Project Deliverables (continued)

Interactive Application (shiny app):

a) A shiny app needs to be created and deployed on shinyapps.io

b) To better understand your model's energy prediction

c) To better understand the potential future energy needs and/or savings

Expectations

1) Work at a consistent pace throughout the rest of the semester
2) Tasks should be distributed equally across the team members
3) Tasks should typically not take a long time to complete – one week target, two weeks is fine, but not a month
4) Tasks should be at an appropriate level of effort/detail

Project Grading

Presentation (5%)

0.5% - Business Questions – Describe the goals of the project

0.5% - Use of Descriptive statistics - Provide context and a basic understanding of the data

1% - Use of modeling techniques – Show the results of different models and explain why they were/were not useful

1% - Visualization - Convey the results in an easy-to-understand manner

1% - Interpretation of the results/Actionable Insights – Make sure the results are actionable (as compared to just interesting)

1% - Know your audience - Present findings in an easy-to-understand way (ex. no data science lingo, easy for others to follow the logic)

Shiny App (5%)

1% - App can load/use a data file

1% - Display of the first ‘n’ rows of the read in dataset

1% - Display visualizations of key drivers of energy usage

1% - Generate predictions via a stored model

1% - An explanation within the app of how to interpret the model outputs

Word Document (15%)

1% - Business Questions - Describe the goals of the project

1% - Data cleanse/munge/preparation - Transform/clean/munge the data appropriately, including missing values

1% - Use of Descriptive statistics - Provide context and a basic understanding of the data

4% - Use of modeling techniques - Try at least 3 different models and compare results

3% - Visualization - Convey information in an easy-to-understand manner

4% - Interpretation of the results/Actionable Insights – Make sure the results are actionable (as compared to just interesting)

1% - Validation - How do you know your results were correct (i.e., no errors)?

Final Project (Word doc)

Example Table of Contents:

- Introduction (scope/context/background)
- Business Questions addressed
- Data Acquisition, Cleansing, Transformation, Munging
- Descriptive statistics & Visualizations
- Use of modeling techniques & Visualizations
  (noting techniques explored but not used in presentation)
- Actionable Insights / Overall interpretation of results